

Effects of Artificial Landmarks on Spatial Learning in FastTap Menus with Large Command Sets

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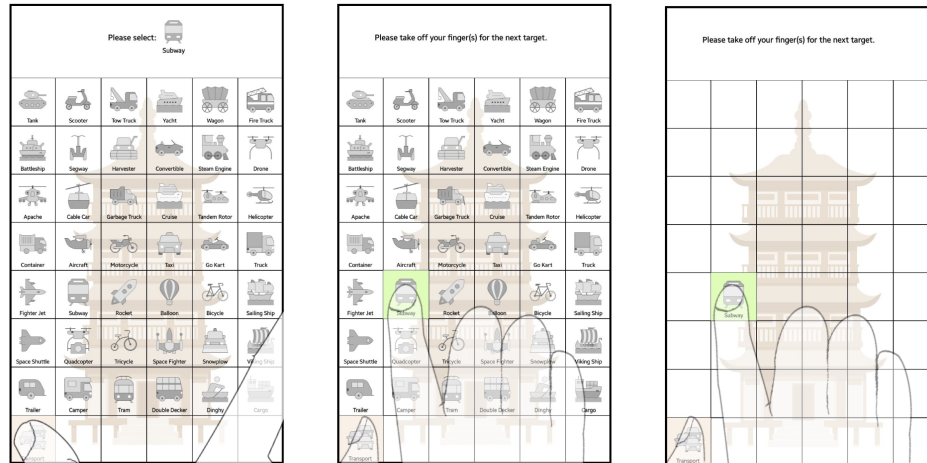


Figure 1: Command selections with FastTap: From left to right: menu invocation - open the menu by pressing the menu button with the thumb; novice selection - select a command (index) after invoking the menu (thumb); expert selection - press the menu button (thumb) and correct command location (index) simultaneously without invoking the entire menu.

Abstract

Spatially-stable menus, FastTap, use grids to display commands on touch tablets, where grid lines act as landmarks helping users quickly learn command locations. While these menus can increase the grid size to allow more commands, the repetitive lines may weaken landmark support and hinder spatial learning. Adding landmarks to desktop menus can help users learn command locations, but it is unclear if this will be effective on tablets with many commands. Therefore, we conducted a study with different landmarking strategies (standard, menu-backdrop image, and coloured blocks) in FastTap menus with large command sets. Results indicated that people developed spatial memory of commands in all interfaces. Once learned, coloured blocks substantially improved command selection performance over the standard and image. People also issued significantly more expert selections with coloured blocks. We demonstrate that artificial landmarks can aid spatial learning in

tablets with large command sets, which will help the future design of tablets.

CCS Concepts

• Human-centered computing → Empirical studies in HCI.

Keywords

Command selection, landmark, spatial memory, tablet, FastTap

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1 Introduction

Placing command icons in fixed locations on a computer display enhances user familiarity with Graphical User Interfaces (GUIs). Consistent positioning allows users to quickly predict where to find commands. Researchers have used spatial consistency to design better interfaces. For example, Marking Menus [22] enabled rapid command selections among a large command set through spatial gestures. Scarr et al.'s CommandMaps [30] demonstrated that spatially stable menu layouts offer superior performance benefits over traditional hierarchical menus in desktops, such as ribbon menus.

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However, the advantages of spatially stable interfaces depend on the user's ability to memorize command locations, which can be challenging due to the presence of a large number of commands in GUIs.

The portability and prevalence of tablet devices have made them a compelling platform for investigating spatially stable GUIs. For example, Schramm et al. [33] placed a hidden toolbar along the bezels of tablets. Gutwin et al.'s FastTap Menu [16] employs a spatially stable grid layout to enable rapid command selection on tablets. Although these spatial-memory-based menus show performance benefits over Marking Menus [22], they can accommodate only a limited number of commands in their interfaces. For example, FastTap uses a stable 5x4 grid to display 19 items, and the hidden toolbar menu has a capacity of 28 items, which is unsuitable to accommodate the large command vocabulary of modern GUIs (Microsoft Office and Adobe Photoshop have hundreds of commands).

One way to increase the command capacity of spatial memory menus like FastTap is by adding more columns and rows to the existing grid. However, increasing the grid size of FastTap menus to accommodate more items can significantly impede spatial learning and recall [1–3]. Therefore, an important challenge remains in determining effective ways to support users in learning command locations on tablets, particularly with many commands.

An effective method to enhance spatial learning is inspired by how people use landmarks to improve their spatial memory. Landmarks are easily identifiable features in space that stand out from their surroundings [24]. Uddin et al. [38] explored artificially developed objects such as an image as a menu background and colour blocks as landmarks in desktop GUIs. They showed that landmarks can effectively help with spatial learning and improve user interaction, particularly in larger interfaces. While adding landmarks to tablets may aid spatial learning, it is uncertain how adding extra objects to smaller interfaces affects the learning of command locations.

We, therefore, conducted a controlled lab study using the single-tab FastTap Menus with an 8x6 grid layout by adapting the FastTap Menu [16]. So, we had 42 commands in the menu. We explored three types of landmarking strategies (adapted from [38]) in the study: a standard grid, an image as the grid's background, and coloured blocks in the grid.

Results indicated that all interfaces supported spatial memory development. However, the spatial benefits were significant when simple and abstract landmarks, coloured blocks, were present in the interface. Despite providing rich and semantically meaningful landmarking features, the image landmark was the least effective landmark in our study. Participants also favoured the coloured blocks as landmarks.

Our research enhances the understanding of spatial interface design in several ways. First, this is the first study to investigate the effects of artificial landmarks on spatial learning in tablets with a large command set. Second, we offer new empirical evidence showing that simple landmarks not only enhance performance by reducing completion times and error rates but also help users achieve expert-level proficiency more quickly. Third, our results indicate that integrating simple landmarks into spatial tablet interfaces, such as the FastTap Menu, can significantly enhance user experience, reduce cognitive load, and improve task efficiency. Last,

these findings will assist designers in creating spatial memory-based interfaces enabling rapid command selection, especially those with larger command vocabularies.

2 Related Work

2.1 Spatial Memory-based Interfaces

Spatial memory has been exploited in GUIs to enable rapid selection [10, 29, 32], ultimately helping the users to become experts [7, 43]. For example, Scarr et al.'s CommandMaps [30, 31], present commands in stable locations in desktop GUIs, facilitating a faster command recall. It was possible because users could leverage spatial knowledge to recall the locations of commands from memory [8, 30, 38]. Later, Gutwin et al.'s ListMaps [14, 15] and Cockburn et al. [6] showed that instead of a linear menu, a stable, flat, and all-item-at-once menu layout can improve item revisitation speed. Researchers have also exploited spatial memory to improve linear document revisitation performance [4, 6, 26, 39].

Besides desktop interfaces, spatial memory can benefit interactions in multi-touch devices as well. For example, Gutwin et al.'s FastTap Menus [16] utilizes a stable flat grid menu to display everything together on tablets, enabling a rapid command selection. Researchers demonstrated that FastTap menus could be multiplexed by adding four tabs without significantly compromising selection speed [12]. Similarly, spatial memory GUIs were shown to improve selection speed in tablets [15, 16, 36], smartwatches [23], smartphones [41, 42], and tabletops [40]. Further studies suggest spatial memory can be useful in Large environments (e.g., VR [11], wall display [19], and tables [20]). However, no tablet menu has explored the effects of landmarks with large command sets.

2.2 Artificial Landmarks in GUIs

Landmarks exist in nature, aiding us to remember a route using a signboard, a specific tree, or how animals navigate back to their homes [5, 25]. Landmarks are distinctive visual features that help users recall locations [24]; and as in real-world navigation, landmarks have shown potential in computer interfaces (e.g., [33, 35, 38]).

In graphical menus, people often use existing landmarks to find commands [36]. For example, the grid lines of the FastTap menu [16], the bezels of the tablet and smartwatch [23, 33], and the corners [16] can serve as landmarks and help with spatial learning, especially in smaller menus. However, as the number of items increases, these landmarks may become less effective, as one landmark may be used as a reference point for multiple items. In these situations, artificially implanted landmarks such as images or simple visual anchors can be effective [38]. Even the user's own hands and fingers can be landmarks on touch surfaces [36, 40]. Researchers explored landmarks in GUIs to enhance revisitation in documents [26, 39], improve multi-touch interactions [34] and to enhance selection performance in spatial memory interface [37, 38]. However, those landmarks were tested on either desktops or large touch interfaces; we know little if those landmarks will work on tablets as they have limited space.

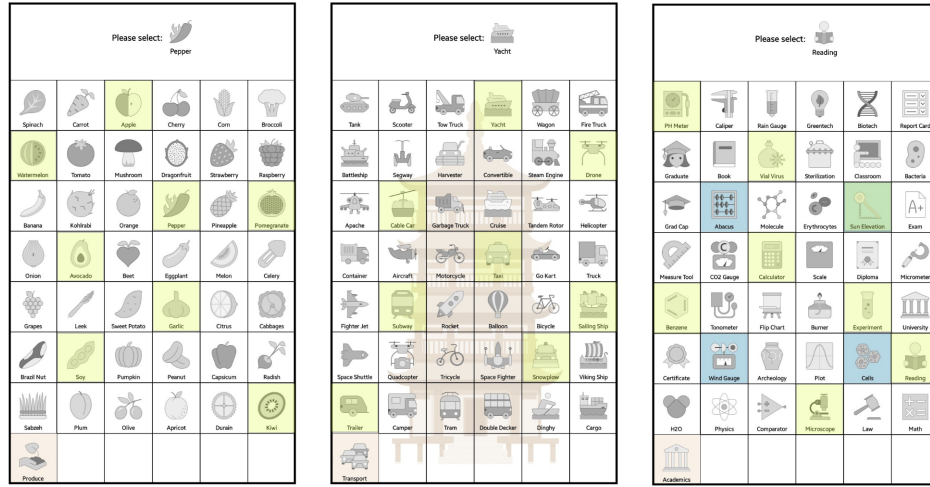


Figure 2: Interfaces used in the Study with highlighted target locations. Images from left to right: Standard interface with only a grid, Image interface with a Japanese Shrine’s image as a backdrop, and Anchor interface with four teal-coloured blocks.

3 Interfaces to Test Landmarks in Tablets

To test the effect of artificial landmarks on spatial learning of commands in tablets with large command sets, we developed three similar interfaces (Figure 2) adapting FastTap Menus [16]. Fast-Tap is a spatial memory-based rapid command selection method that uses the entire tablet screen to display all commands simultaneously in a fixed grid layout. By default, the commands remain hidden, users can only access them by pressing and holding a menu button (appears at the bottom-left grid cell) with their thumb. Commands can be selected using an index finger while the menu button remains pressed. As users learn command locations in the grid, they can predict the location where a command will be displayed, enabling rapid selections, recalling locations from memory. Early research with FastTap tested its performance benefits with a very limited number of commands, only 19 items in a 5x4 grid [16].

Since our focus was on larger command sets in tablets, we needed to increase the command capacity of the FastTap menu. In an effort to explore spatial learning in FastTap menus with large command sets, Abdullah et al. designed FastTap menus in four sizes: Small (5x4), Medium (7x5), Large (8x6), and Extra-Large (9x7) [1, 2] and in a subsequent study, they introduced a multi-tab FastTap menu with Small (5x4), Medium (7x5), and Large (8x6) grid sizes, excluding the Extra-Large size due to its overwhelming nature for participants [3]. Their research showed that increasing the menu size significantly impaired the development of spatial memory. Therefore, we chose the Large (8x6) grid size for our study, striking a balance between increased capacity and manageable selection performance. We used an 8.7-inch Samsung Galaxy Tab A7 Lite tablet for our study. We kept the menu operation identical to the original FastTap menu [16] and considered Parhi et al.’s [27] recommendation for touch target size (at least 9.6mm). We kept about 1.7 inches of space at the top of the grid to display target questions. Therefore, the grid was approximately 7 inches diagonally, matching the original FastTap design. The bottom row was reserved for menu buttons. So, we had 42 items in the menu.

As shown in Figure 2, our three testing interfaces initially display only a blank grid (Standard, left), a grid along with a transparent image of a Japanese Shrine (Image, middle), and a grid with four teal-coloured blocks (Anchor, right). Upon pressing the menu button (bottom-left corner cell), commands were displayed by overlaying them on the grid, which can be selected with an index finger.

In this work, we adapted the three landmarking strategies (standard grid, menu-backdrop image, and coloured blocks) from Uddin et al. [38]. The standard grid provides clear intersecting lines along with inherent referencing points (e.g., bezels and corners) that may help to learn command locations on tablets. However, when the grid size increases, the existing landmarks and repetitive lines become weak. In such cases, an image as the menu backdrop and four coloured blocks may provide learning benefits with concrete and abstract objects, respectively. While the image provides rich, semantically meaningful landmarking features, the coloured blocks provide simple and abstract landmark features.

Our three interfaces supported expert and novice ways of selection. We implemented a 200ms delay between pressing the menu button and the icons appear. If the user was familiar with the menu, they could select a command by pressing the target location and menu simultaneously after recalling its location from memory: an expert selection. Otherwise, they could wait for the entire grid to appear, visually search for the target, and then make a selection, which is considered a novice selection. The landmarks were always present in both selection modes (Figure 1).

4 Study: Artificial Landmarks in FastTap Menus

The study was designed to explore *whether and how different artificial landmarks affect spatial learning in tablets with large command sets*.

4.1 Study Method

4.1.1 Tasks and Stimulus. The study involved trials of command selections where the target (image and text) was displayed at the

top of the grid. Trials began when the target question appeared on the screen. Users could select the target using their index finger while simultaneously pressing the menu activation button with their thumb. If the user selected the correct target, the background of the grid cell turned green; otherwise, it turned red for feedback. After a successful selection, the user must remove their thumb from the menu to see the next question.

There were three conditions (Standard, Image and Anchor), each containing 42 grayscale commands from produce, vehicle and educational categories, respectively. Each condition had 18 blocks, with three blind blocks appearing every sixth block, where the menu icons remained hidden even after pressing the menu button. This blind block encourages participants to rely on their memory by remembering target positions. Eight targets were quasi-randomly selected as stimuli from each interface; none were reused in subsequent interface conditions.

4.1.2 Procedure and Study Design. We conducted a within-subject study with three interface conditions: standard, image and anchor. There were eighteen blocks in each condition, including three blind blocks. Every block consisted of the same eight trials (order randomized), and the order of the conditions was selected using a Latin Square design [13].

Participants were informed that the study was concerned with rapid selection. Expert and novice ways of selection were described and demonstrated. They then practiced on a 4x3 grid (2 targets over 10 blocks) with a different command set to become familiar with menu operations. Participants were instructed to complete tasks as fast and accurately as possible. They completed a NASA-TLX [17] questionnaire after each condition and provided preferences after completing all conditions.

4.1.3 Participants and Apparatus. Eighteen individuals (nine Cis men, seven Cis women, one Trans man, and one prefer not to say; one left-handed; aged between 19 and 32; mean age 23.56) were recruited from a local university to participate in the study. All participants were familiar with touch devices, with an average of 9.72 years of experience, and an average of 5.78 hours daily usage. The study session lasted approximately 60 minutes, and each participant received a \$10 gift card as an honorarium.

The experiment was carried out on an 8.7-inch multi-touch display of a Samsung Galaxy Tab A7 Lite tablet. The application was developed for the Android platform using Java.

4.2 Results

Our samples followed an approximately normal distribution as per Shapiro-Wilk tests, which were cross-checked with Q-Q plot analyses. Therefore, we proceeded with ANOVA for our analyses. We calculated the effect size for significant RM-ANOVA results as general eta-squared: η^2 (considering 0.01 as small, 0.06 as medium, and greater than 0.14 as large [9]). We applied Bonferroni correction for post-hoc pairwise t-tests.

4.2.1 Trial Completion Time. Trial completion time was measured from the display of the target to its accurate selection. Mean completion times are summarized for regular and blind blocks (B6, B12, and B18) in Figure 3.

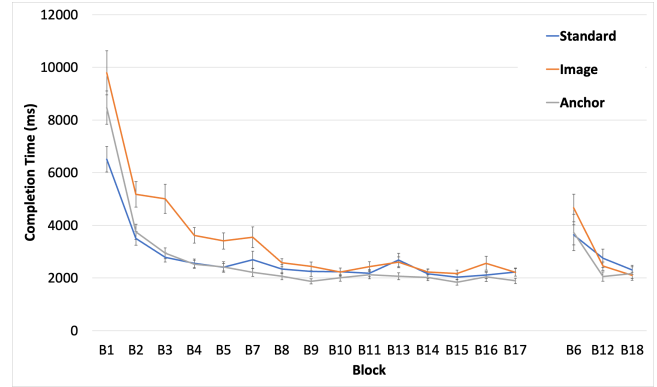


Figure 3: Completion Time (±s.e.) by Condition and Block

ANOVA found a main effect of condition ($F_{2,34} = 17.93, p < 0.001, \eta^2 = 0.09$) for regular blocks, with Anchor (mean 2682.27, $sd = 2236.56$) outperforming Standard (2710.33, $sd = 2431.76$) and Image (3466.26, $sd = 3494.07$). Trial time significantly decreased over regular blocks ($F_{14,238} = 63.92, p < 0.001, \eta^2 = 0.63$), and there was a significant condition $\times$ block ($F_{28,476} = 4.54, p < 0.001, \eta^2 = 0.11$) interaction. Follow-up tests also showed significant differences between Standard:Image and Image:Anchor pairs (all $p < 0.001$), indicating Anchors provided better spatial learning support.

For blind blocks, we found a significant effect of block ($F_{2,34} = 14.42, p < 0.001, \eta^2 = 0.27$) as trial time decreased over time, but no effect on condition ($F_{2,34} = 1.24, p = 0.30$) and condition $\times$ block interaction ($F_{4,68} = 1.99, p = 0.11$).

4.2.2 Expert Selection. The expert selection occurs when a participant chooses a target command without invoking the menu, regardless of accuracy, in less than 200ms. Figure 4 summarizes these results for fifteen regular blocks.

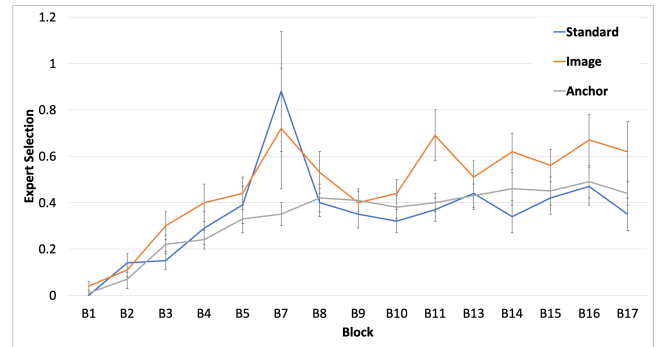


Figure 4: Expert Selections (±s.e.) by Condition and Block, where commands were selected without invoking the menu.

Interestingly, in Image (mean 0.47, $sd = 1.03$), participants used more expert selections than Standard (mean 0.35, $sd = 0.87$) and Anchor (mean 0.34, $sd = 0.60$). However, ANOVA revealed no significant effect of condition ($F_{2,34} = 2.74, p = 0.08$) and condition $\times$ block ($F_{28,476} = 1.24, p = 0.18$), but a significant effect was found

on block ($F_{14,238} = 4.58$, $p < 0.001$, $\eta^2 = 0.12$) as expert selection dramatically increased over time.

4.2.3 Errors in Selection. The error count for each trial was calculated based on the number of incorrect targets selected before the correct one. For regular blocks, errors did not increase over time. Thus, ANOVA found no significant effect of block ($F_{14,238} = 1.36$, $p=0.17$) but revealed significant effects in condition ($F_{2,34} = 6.71$, $p=0.003$, $\eta^2 = 0.03$) and condition $\times$ block ($F_{28,476} = 1.58$, $p=0.03$, $\eta^2 = 0.04$). Participants made fewer errors on the Anchor (mean 0.09, $sd = 0.40$) than Standard (mean 0.19, $sd = 0.74$) and Image (mean 0.24, $sd = 1.06$). Post-hoc tests indicated significant differences between all conditions ($p<0.003$) except for the Standard and Image positions ($p=0.09$).

For blind blocks, ANOVA showed a significant effect of block ($F_{2,34} = 11.11$, $p<0.001$, $\eta^2=0.18$) but no effect for condition ($F_{2,34} = 0.29$, $p=0.75$) and condition $\times$ block ($F_{4,68} = 1.51$, $p=0.20$).

4.2.4 Subjective Responses. We analyzed the raw NASA-TLX responses and found a significant difference among all the conditions (all $p < 0.05$), and the Anchor condition had the lowest workload scores with the best performance according to participants. Post-hoc tests found no differences for any measures.

Table 1: Participant preferences on conditions in terms of different factors.

	Standard	Image	Anchor
Speed	1	6	11
Accuracy	3	5	10
Memorization	1	6	11
Expert Mode	3	5	10
Comfort	5	5	8
Overall	3	5	10

Participant preferences are outlined in Table 1. It shows that in all categories, both the Anchor and Image landmarked conditions received higher preferences among participants. However, most participants favoured the Anchor condition. This indicates that participants found the landmarks in the interfaces beneficial.

5 Discussions

5.1 Explanations for Results

5.1.1 Did Landmarks Facilitate Spatial Learning? In our study, the mean trial completion time significantly decreased across blocks, irrespective of landmarking strategies (see Figure 3). These reductions followed the power law of the learning curve [18] (all $R^2 > 0.8$), as expected. This indicates that our participants shifted from slow visual search selections to rapid selections based on memory. These transitions occurred in all interfaces, indicating that spatial memory development took place in all interfaces. However, the Anchor condition, consisting of coloured blocks within the grid, provided spatial cues to memorize command locations more effectively. As a result, it led to faster trial completion times, higher expert selections, and fewer errors compared to other conditions. It suggests that participants in our study could effectively develop memories

of command locations and later exploit those more quickly in interfaces with large command sets when additional referencing mechanisms (i.e., landmarks) were present in interfaces, particularly in the anchor condition.

5.1.2 Why Coloured Blocks Were More Effective in Supporting Spatial Learning? The Anchor condition with simple and abstract coloured blocks showed significantly better user performance compared to the Image in our study, which aligns with prior research [38]. However, it contradicts prior evidence from a popular memory enhancement mechanism, the ‘method of loci’ [28], suggested that semantically meaningful referencing points, such as physical structures, can provide better memorization aid.

We believe there are two main reasons behind the Anchor’s superior performance. First, we used a two-dimensional representation mechanism, a grid, to present commands in our designs. All lines of the grid were straight and provided an ordered structure. The Anchor condition used coloured grid cells, which matched the grid structure. This strong alignment between the grid and the coloured blocks might have been easy to exploit to remember target locations. For example, “the target was one or two cells away from the block”, whereas, in the Image condition, the structure or shape of the image misaligned from the grid. It may have been difficult for the participants to form clear connections between the target and the image.

Second, the image landmark was composed of numerous intricate features, such as doors, windows, floor, roof, etc., whereas the Anchor consisted of simple coloured blocks evenly spaced in the grid. Research also indicates that greater visual complexity in web designs often creates more positive initial impressions among users and improves memorability [21]. However, our findings suggest that the complex design of the Image condition may have distracted users during their interaction with a large command set in a tablet menu. As a result, even the only grid line-based Standard interface, along with its inherent landmarking features (lines, corners, or edges), provided better support for spatial learning than the Image in our study. A simpler image could yield better landmarking support.

5.2 Design Implications and Generalizing the Results

Our results showed that integrating simple and abstract landmarks into the FastTap Menu with large command sets significantly improves user performance, suggesting that designers could consider adding such landmarks when designing spatial memory-based tablet GUIs with many commands to enhance spatial learning and recall.

FastTap menus with artificial landmarks have strong potential for real-world applications, particularly with numerous commands. Applications like Adobe Illustrator and Microsoft PowerPoint can use FastTap menus on tablets to manage extensive command sets. By adding landmarks, designers help users reduce visual search time and enhance workflow efficiency. Furthermore, these techniques hold potential in designing applications in mobile gaming, assistive technologies, and AR/VR interfaces, where quick navigation among a large number of contents is crucial.

5.3 Limitations and Future Work

Our research offers valuable insights into spatial learning in tablets, but there are some limitations and areas for future exploration. One limitation is that we focused on only two types of landmarks: images as menu-backdrop and anchors. Future studies could explore a wider range of landmark types to determine potential performance benefits. Additionally, combining multiple landmark types in one interface may enhance user performance and satisfaction further.

We used different sets of icons in our study (see Figure 2). The learning performance may be affected by the familiarity of icon sets among participants. For example, Education icons may be more challenging to recognize than Produce icons, or vice versa. Future studies could investigate the impact of icon familiarity on spatial learning of commands.

6 Conclusions

We presented a controlled lab study to explore the effectiveness of artificial landmarks on spatial learning with large command sets in FastTap Menus. We used an 8x6 grid layout with three landmarks – standard grid, menu backdrop image, and colour blocks as anchors to test user learning. Our findings indicated that using additional simple but abstract landmarks in tablets could significantly enhance spatial learning when a large command set was present. We contribute empirical evidence that can lay the groundwork for incorporating artificial landmarks in designing future touch tablet interfaces featuring large command sets, especially those relying on spatial memory.

Acknowledgments

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References

- [1] Sayeem Md. Abdullah and Md. Sami Uddin. 2024. Effects of Increasing Command Capacity of Spatial Memory Menus in Tablets. In *Companion Proceedings of the 2024 Conference on Interactive Surfaces and Spaces* (Vancouver, BC, Canada) (ISS Companion '24). Association for Computing Machinery, New York, NY, USA, 69–72. <https://doi.org/10.1145/3696762.3698055>
- [2] Sayeem Md Abdullah and Md. Sami Uddin. 2024. Revisiting FastTap: Effects of Increasing Command Capacity of Spatial Memory Menus in Tablets. In *2024 IEEE 15th Annual Ubiquitous Computing, Electronics & Mobile Communication Conference (UEMCON)*. 0683–0689. <https://doi.org/10.1109/UEMCON62879.2024.10754660>
- [3] Sayeem Md Abdullah and Md. Sami Uddin. 2025. Effects of Overloading Command Capacity of Multiplexed FastTap Menus on Spatial Learning in Tablets. In *Proceedings of the 20th International Joint Conference on Computer Vision, Imaging and Computer Graphics Theory and Applications - Volume 1: GRAPP, HUCAPP and IVAPP*. 495–505.
- [4] Jason Alexander, Andy Cockburn, Stephen Fitchett, Carl Gutwin, and Saul Greenberg. 2009. Revisiting read wear: analysis, design, and evaluation of a footprints scrollbar. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems - CHI '09*. ACM, New York, NY, USA, 1665–1674. <https://doi.org/10.1145/1518701.1518957>
- [5] B. A. Cartwright and T. S. Collett. 1982. How honey bees use landmarks to guide their return to a food source. *Nature* 295, 5850 (1982), 560–564. <https://doi.org/10.1038/295560A0>
- [6] Andy Cockburn, Carl Gutwin, and Jason Alexander. 2006. Faster document navigation with space-filling thumbnails. In *Proceedings of the SIGCHI conference on Human Factors in computing systems - CHI '06*. ACM, New York, NY, USA, 1–10. <https://doi.org/10.1145/1124772.1124774>
- [7] Andy Cockburn, Carl Gutwin, Joey Scarr, and Sylvain Malacria. 2014. Supporting Novice to Expert Transitions in User Interfaces. *Comput. Surveys* 47, 2 (Nov. 2014), 1–36. <https://doi.org/10.1145/2659796>
- [8] Andy Cockburn, Per Ola Kristensson, Jason Alexander, and Shumin Zhai. 2007. Hard lessons: effort-inducing interfaces benefit spatial learning. In *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '07*. ACM Press, New York, New York, USA, New York, USA, 1571–1580. <https://doi.org/10.1145/1240624.1240863>
- [9] Jacob Cohen. 1973. Eta-squared and partial eta-squared in communication science. *Human Communication Research* 28, 56 (1973), 473–490.
- [10] Brian D. Ehret. 2002. Learning where to look: location learning in graphical user interfaces. In *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '02*. ACM, New York, NY, USA, 211–218. <https://doi.org/10.1145/503376.503414>
- [11] BoYu Gao, Byungmoon Kim, Jee-In Kim, and HyungSeok Kim. 2019. Amphitheater Layout with Egocentric Distance-Based Item Sizing and Landmarks for Browsing in Virtual Reality. *International Journal of Human-Computer Interaction* 35, 10 (June 2019), 831–845. <https://doi.org/10.1080/10447318.2018.1498654>
- [12] Varun Gaur, Md. Sami Uddin, and Carl Gutwin. 2018. Multiplexing spatial memory: increasing the capacity of FastTap menus with multiple tabs. In *Proceedings of the 20th International Conference on Human-Computer Interaction with Mobile Devices and Services - MobileHCI '18*. ACM Press, New York, New York, USA, 1–13. <https://doi.org/10.1145/3229434.3229482>
- [13] David A. Grant. 1948. The latin square principle in the design and analysis of psychological experiments. *Psychological bulletin* 45, 5 (Sept. 1948), 427–442. <https://doi.org/10.1037/H0053912>
- [14] Carl Gutwin and Andy Cockburn. 2006. Improving list revisitation with ListMaps. *Proceedings of the Workshop on Advanced Visual Interfaces 2006* (2006), 396–403. <https://doi.org/10.1145/1133265.1133347>
- [15] Carl Gutwin, Andy Cockburn, and Benjamin Lafreniere. 2015. Testing the rehearsal hypothesis with two FastTap interfaces. In *Proceedings of the 41st Graphics Interface Conference - GI '15*. Canadian Information Processing Society, 223–231.
- [16] Carl Gutwin, Andy Cockburn, Joey Scarr, Sylvain Malacria, and Scott C. Olson. 2014. Faster command selection on tablets with FastTap. In *Proceedings of the ACM conference on Human factors in computing systems - CHI '14*. ACM Press, New York, New York, USA, 2617–2626. <https://doi.org/10.1145/2556288.2557136>
- [17] Sandra G. Hart and Lowell E. Staveland. 1988. Development of NASA-TLX (Task Load Index): Results of Empirical and Theoretical Research. *Advances in Psychology* 52, C (Jan. 1988), 139–183. [https://doi.org/10.1016/S0166-4115\(08\)62386-9](https://doi.org/10.1016/S0166-4115(08)62386-9)
- [18] William C. Hill, James D. Hollan, Dave Wroblewski, and Tim McCandless. 1992. Edit wear and read wear. In *Proceedings of the SIGCHI conference on Human factors in computing systems - CHI '92*. ACM Press, New York, NY, USA, 3–9. <https://doi.org/10.1145/142750.142751>
- [19] Yvonne Jansen, Jonas Schjerlund, and Kasper Hornbæk. 2019. Effects of Locomotion and Visual Overview on Spatial Memory when Interacting with Wall Displays. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI '19*. ACM Press, New York, New York, USA, New York, USA, 1–12. <https://doi.org/10.1145/3290605.3300521>
- [20] Nikhita Joshi and Daniel Vogel. 2019. An Evaluation of Touch Input at the Edge of a Table. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI '19*. ACM Press, New York, New York, USA, 1–12. <https://doi.org/10.1145/3290605.3300476>
- [21] Andy J King, Allison J Lazard, and Shawna R White. 2020. The influence of visual complexity on initial user impressions: Testing the persuasive model of web design. *Behaviour & Information Technology* 39, 5 (2020), 497–510.
- [22] Gordon Kurtenbach and William Buxton. 1994. User learning and performance with marking menus. In *Proceedings of the SIGCHI conference on Human factors in computing systems celebrating interdependence - CHI '94*. ACM Press, New York, New York, USA, 258–264. <https://doi.org/10.1145/191666.191759>
- [23] Benjamin Lafreniere, Carl Gutwin, Andy Cockburn, and Tovi Grossman. 2016. Faster Command Selection on Touchscreen Watches. In *Proceedings of the ACM Conference on Human Factors in Computing Systems - CHI '16*. ACM Press, New York, New York, USA, 4663–4674. <https://doi.org/10.1145/2858036.2858166>
- [24] Kevin Lynch. 1960. *The image of the city*. MIT Press.
- [25] Pierre Emmanuel Michon and Michel Denis. 2001. When and Why Are Visual Landmarks Used in Giving Directions? *Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)* 2205 (2001), 292–305. https://doi.org/10.1007/3-540-45424-1_20
- [26] Ehsan Sotoodeh Mollashahi, Md. Sami Uddin, and Carl Gutwin. 2018. Improving revisitation in long documents with two-level artificial-landmark scrollbars. In *Proceedings of the 2018 International Conference on Advanced Visual Interfaces*. ACM, New York, NY, USA, 1–9. <https://doi.org/10.1145/3206505.3206554>
- [27] Pekka Parhi, Amy K. Karlson, and Benjamin B. Bederson. 2006. Target size study for one-handed thumb use on small touchscreen devices. *ACM International Conference Proceeding Series* 159 (2006), 203–210. <https://doi.org/10.1145/1152215.1152260>
- [28] Simon T. Perrault, Eric Lecolinet, Yoann Pascal Bourse, Shengdong Zhao, and Yves Guiard. 2015. Physical Loci: leveraging spatial, object and semantic memory for command selection. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems - CHI '15*. ACM, New York, NY, USA, 299–308. <https://doi.org/10.1145/2659796>

- [//doi.org/10.1145/2702123.2702126](https://doi.org/10.1145/2702123.2702126)
- [29] George Robertson, Mary Czerwinski, Kevin Larson, Daniel C. Robbins, David Thiel, and Maarten van Dantzich. 1998. Data mountain: using spatial memory for document management. In *Proceedings of the ACM symposium on User interface software and technology - UIST '98*. ACM Press, New York, New York, USA, 153–162. <https://doi.org/10.1145/288392.288596>
 - [30] Joey Scarr, Andy Cockburn, Carl Gutwin, and Andrea Bunt. 2012. Improving command selection with CommandMaps. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems - CHI '12*. ACM, New York, NY, USA, 257–266. <https://doi.org/10.1145/2207676.2207713>
 - [31] Joey Scarr, Andy Cockburn, Carl Gutwin, and Sylvain Malacria. 2013. Testing the robustness and performance of spatially consistent interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. ACM, 3139–3148. <https://doi.org/10.1145/2470654.2466430>
 - [32] Joseph Laurence Scarr. 2014. *Understanding and Exploiting Spatial Memory in the Design of Efficient Command Selection Interfaces*. Ph.D. Dissertation. University of Canterbury.
 - [33] Katherine Schramm, Carl Gutwin, and Andy Cockburn. 2016. Supporting Transitions to Expertise in Hidden Toolbars. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems - CHI '16*. ACM, New York, NY, USA, 4687–4698. <https://doi.org/10.1145/2858036.2858412>
 - [34] Md. Sami Uddin. 2016. *Improving Multi-Touch Interactions Using Hands as Landmarks*. Master's thesis. University of Saskatchewan.
 - [35] Md. Sami Uddin. 2022. *Use of Landmarks to Improve Spatial Learning and Revision in Computer Interfaces*. Ph.D. Dissertation. University of Saskatchewan.
 - [36] Md. Sami Uddin and Carl Gutwin. 2016. Rapid Command Selection on Multi-Touch Tablets with Single-Handed HandMark Menus. In *Proceedings of the 2016 ACM International Conference on Interactive Surfaces and Spaces*. ACM, New York, NY, USA, 205–214. <https://doi.org/10.1145/2992154.2992172>
 - [37] Md. Sami Uddin and Carl Gutwin. 2021. The Image of the Interface: How People Use Landmarks to Develop Spatial Memory of Commands in Graphical Interfaces. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 1–17. <https://doi.org/10.1145/3411764.3445050>
 - [38] Md Sami Uddin, Carl Gutwin, and Andy Cockburn. 2017. The Effects of Artificial Landmarks on Learning and Performance in Spatial-Memory Interfaces. *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (2017). <https://doi.org/10.1145/3025453>
 - [39] Md. Sami Uddin, Carl Gutwin, and Alix Goguy. 2017. Using artificial landmarks to improve revision performance and spatial learning in linear control widgets. In *Proceedings of the 5th Symposium on Spatial User Interaction*. ACM, New York, NY, USA, 48–57. <https://doi.org/10.1145/3131277.3132184>
 - [40] Md. Sami Uddin, Carl Gutwin, and Benjamin Lafreniere. 2016. HandMark Menus: Rapid Command Selection and Large Command Sets on Multi-Touch Displays. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 5836–5848. <https://doi.org/10.1145/2858036.2858211>
 - [41] Shumin Zhai and Per-Ola Kristensson. 2003. Shorthand writing on stylus keyboard. In *Proceedings of the conference on Human factors in computing systems - CHI '03*. ACM Press, New York, New York, USA, 97–104. <https://doi.org/10.1145/642611.642630>
 - [42] Jingjie Zheng, Xiaojun Bi, Kun Li, Yang Li, and Shumin Zhai. 2018. M3 Gesture Menu: Design and Experimental Analyses of Marking Menus for Touchscreen Mobile Interaction. In *Proceedings of the Conference on Human Factors in Computing Systems - CHI '18*. ACM Press, New York, NY, USA, 1–14. <https://doi.org/10.1145/3173574.3173823>
 - [43] Jingjie Zheng, Blaine Lewis, Jeff Avery, and Daniel Vogel. 2018. FingerArc and FingerChord: Supporting Novice to Expert Transitions with Guided Finger-Aware Shortcuts. In *The 31st Annual ACM Symposium on User Interface Software and Technology - UIST '18*. ACM Press, New York, New York, USA, 347–363. <https://doi.org/10.1145/3242587.3242589>